

Myalgias and Musculoskeletal Concerns

Myalgias are the most common musculoskeletal complaint, occurring in 15 percent of patients. In severe cases, creatine phosphokinase levels should be evaluated to assess for possible rhabdomyolysis.

Isotretinoin Therapy: Teratogenicity and Contraception

The greatest concern during isotretinoin therapy is the risk of administration during pregnancy, which can lead to severe teratogenic effects in the fetus. The drug is not mutagenic; its adverse effects occur during organogenesis. Therefore, isotretinoin should be reserved for patients who have not responded to other treatments.

Women of childbearing age must be fully informed of the risks associated with pregnancy while on isotretinoin. Patients must use two highly effective forms of contraception—such as an oral contraceptive combined with condoms and spermicidal jelly—starting at least one month before beginning therapy. Female patients must be thoroughly counseled and demonstrate understanding of contraceptive methods prior to treatment initiation. Dual contraception must continue throughout isotretinoin therapy and for one month after discontinuation.

In the United States, the FDA requires a strict risk management program known as iPledge. This computer-based system ensures adherence to pregnancy prevention protocols before, during, and after isotretinoin treatment.

Dosing and Treatment Duration

The recommended daily dosage of isotretinoin is 0.5 to 1.0 mg/kg/day. A cumulative weight-based dosing target of 120 to 150 mg/kg over the course of therapy is often used, particularly in patients with variable dosing or interrupted treatment, as achieving this total dose improves the likelihood of long-term remission.

Patients with severe truncal acne—especially those with back and chest lesions, which respond less robustly than facial lesions—may require higher doses, up to 2 mg/kg/day.

In patients with severe acne, particularly those with granulomatous lesions, disease flares are common at treatment initiation. To mitigate this, initial dosing should be low—often below 0.5 mg/kg/day. Pre-treatment with prednisone (40 to 60 mg/day) for 1 to 2 weeks may be necessary and sometimes continued for the first two weeks of isotretinoin therapy.

A typical course lasts 20 weeks, but duration is flexible. Therapy may be extended in patients who do not achieve adequate response. Additional improvement may occur for 1 to 2 months after stopping treatment, so complete clearance is not required before discontinuation.

Laboratory Monitoring

Baseline laboratory tests should include a complete blood count and liver function tests. However, the most critical monitoring involves serum triglyceride levels.

- Baseline triglycerides should be measured before starting therapy.
- Repeat testing is recommended at 3 to 4 weeks and again at 6 to 8 weeks.
- If levels are normal at 6 to 8 weeks and no risk factors are present, further testing may not be needed.
- If triglycerides exceed 500 mg/dL, monitoring should be frequent.
- Levels above 700–800 mg/dL warrant interruption of therapy or initiation of lipid-lowering treatment.
- Complications such as eruptive xanthomas or pancreatitis can occur at very high triglyceride levels.

Diet and Acne

Several studies have suggested a potential role for diet in acne development. However, the overall evidence remains inconclusive. The impact of chocolate, sweets, milk, and fatty foods on acne requires further investigation. There is no

strong evidence supporting routine elimination of these foods. Nevertheless, some patients report disease flares after consuming specific foods—particularly chocolate. Given these individual perceptions, it is reasonable to advise patients to avoid dietary items they believe trigger their acne.

Acne Surgery

Acne surgery, involving the removal of comedones and superficial pustules, can accelerate the resolution of individual acne lesions. It was historically a mainstay of acne treatment.